

The Link Layer: MAC sublayer

Lecture 23

Link Layer

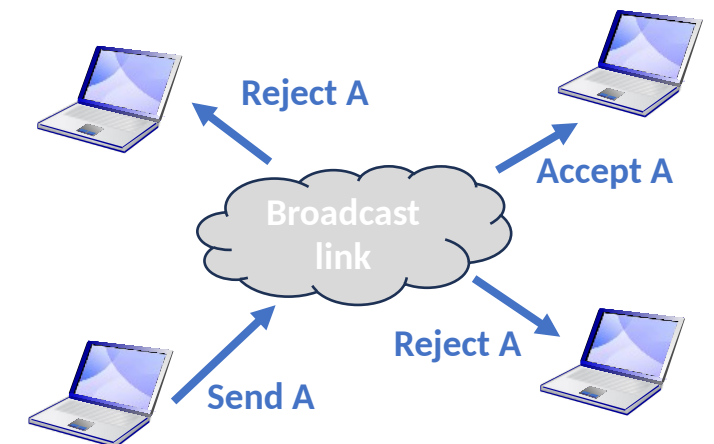
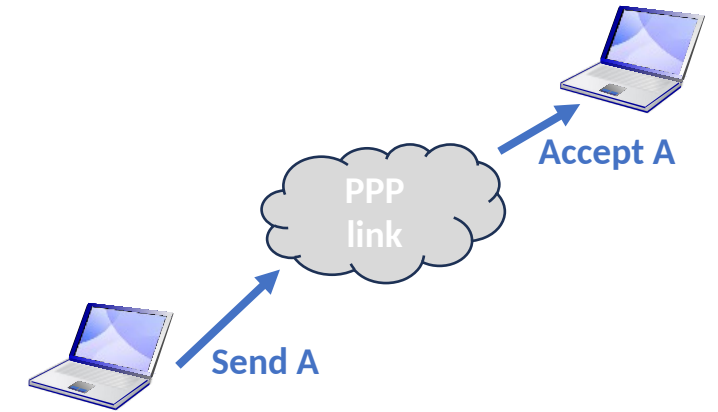
Services

- **Link layer may offer up to 4 services:** framing, link access, reliable delivery, error detection and correction.
 1. **Framing:** to encapsulate/decapsulate datagrams into/from frames.
 2. **Link access:** to regulate access to shared links (broadcast)
 - Done by the **Medium Access Control** (MAC) protocol.
 3. **Reliable delivery:** to guarantee that frames transmitted across the link are received (mostly optional).
 4. **Error detection and correction:** to detect frame's errors and possibly correct them.
 - Parity check, checksum, CRC.

Link Layer

Link Types

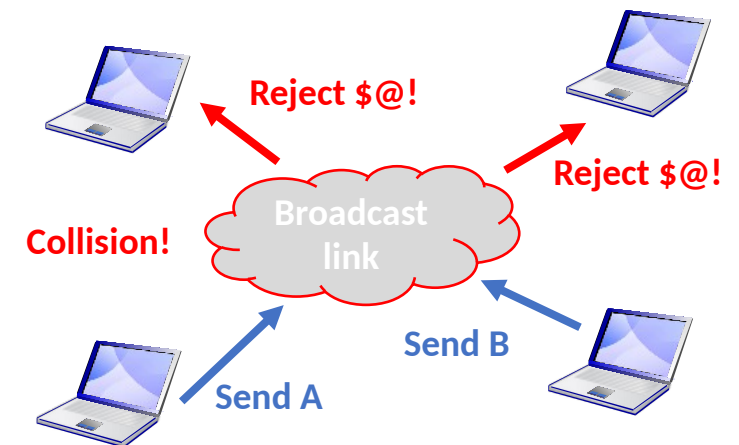
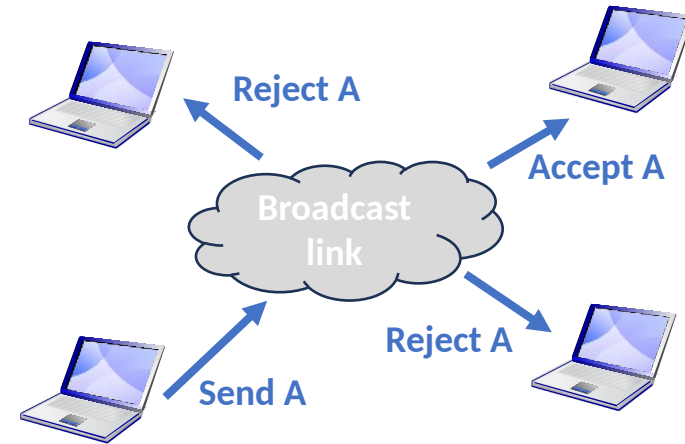
- There are basically 2 types of links that can be managed:
 - **Point-to-point link:** consists of a **single sender** at one end and a **single receiver** at the other end. The **point-to-point protocol** (PPP) is one example of protocol managing such links.
 - E.g., direct ethernet link between 2 computers.
 - **Broadcast link:** multiple sending and receiving nodes all connected to **the same shared channel**. The term broadcast is used because when one node transmits a frame it is received by all nodes on the channel.
 - E.g., Ethernet bus, half-duplex Ethernet (rare, as most cables are today's full-duplex) or wireless LANs.
- The access to **broadcast links have to be coordinated** (multiple access problem) as multiple communication on a single link may interfere each other.



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Collisions

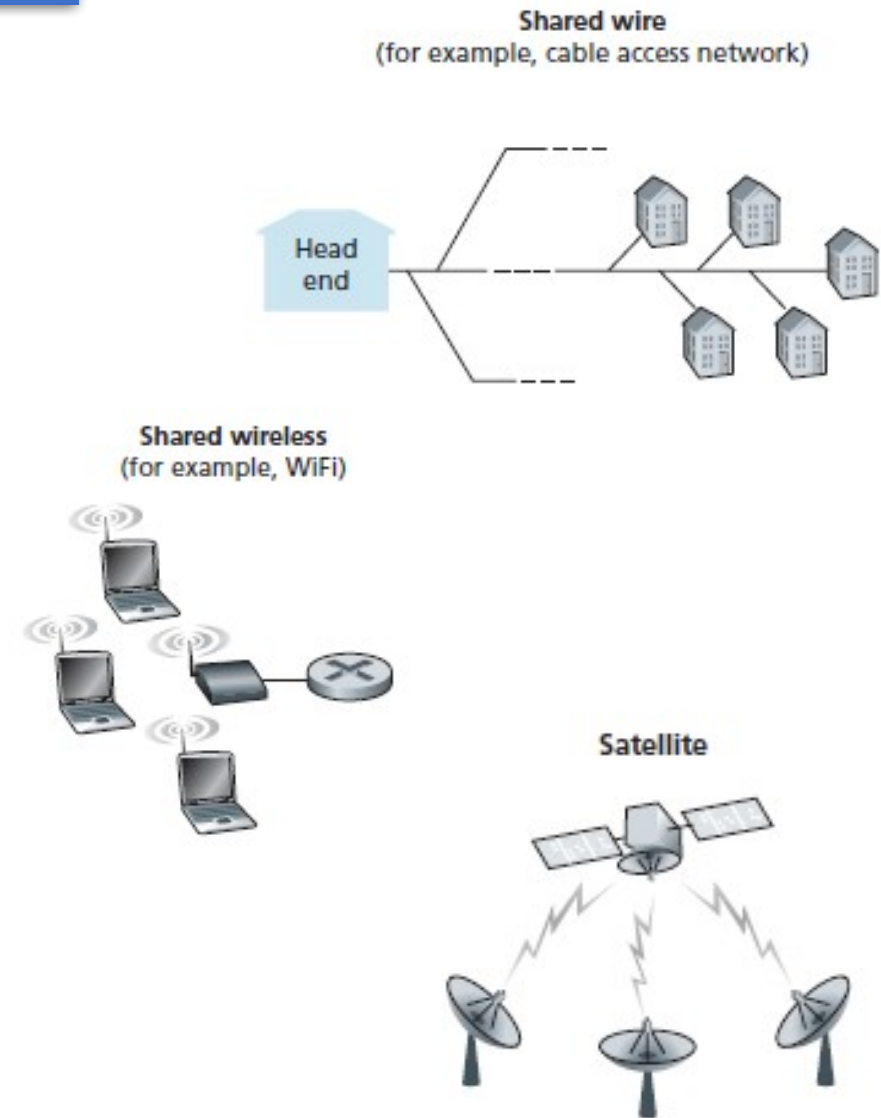
- The main problem of broadcast link is the **collision**: if multiple nodes are simultaneously transmitting frames on the same channel, **all such frames overlap becoming unintelligible**.
- These **collided frames** are then received by all nodes on the channel and **dismissed as errors** (no harm is done), but:
 - All **transmitted frames are lost**.
 - The **time-interval is wasted**, as the channel has been used to transmit useless data.
- The **access** to the physical medium (as well as **framing**) is regulated by the **MAC** (**Medium Access Control**) protocol.



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Multiple Access Protocol

- In computer networks **multiple access protocols** are used **to regulate transmissions via broadcast channels**, so that:
 - **Collisions are managed.**
 - Each node has a chance to transmit, so **nodes do not monopolize the link.**
 - Established **connections are not interrupted** (e.g., checking if link is busy).
- Such protocols are needed for a variety of network settings, including both wired, wireless or satellite networks, **where hundreds or thousands of nodes can directly communicate over a broadcast channel.**



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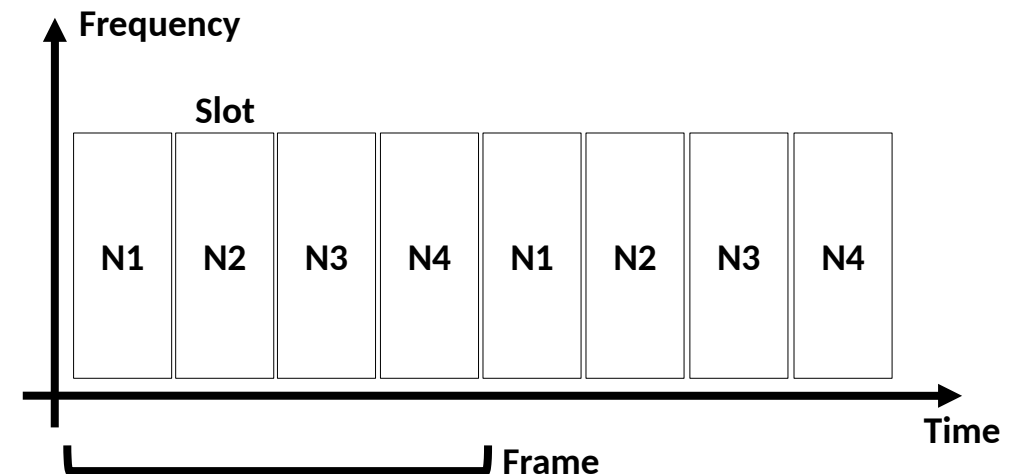
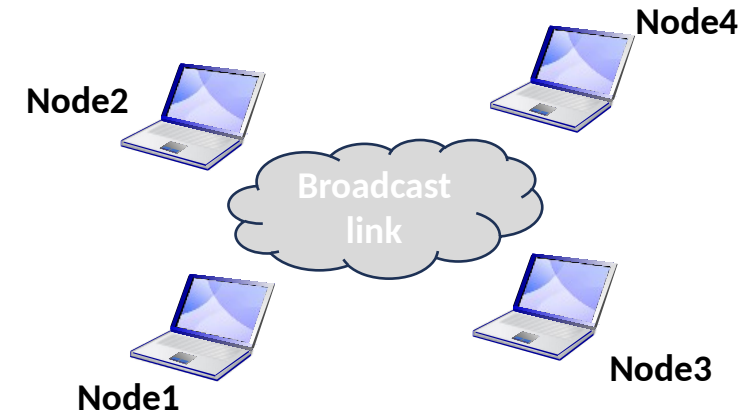
Multiple Access Protocol

- The primary role of a **multiple access protocol** is to somehow **avoid collisions** (there are dozens of protocols over different link-layer technologies). Main approaches are:
 - **Channel partitioning**: the bandwidth is partitioned for different nodes.
 - **Random access**: the nodes “gamble” for the access.
 - **Taking-turns**: the nodes wait for their turn.
- **Desiderata**: a multiple access protocol for a **broadcast channel of rate R bps** should also provide the following characteristics:
 - To **maximize the usage of the channel**: if M nodes have data to send, each one should have, in average, a throughput of R/M bps (if $M=1$ then throughput should be R).
 - To be **decentralized**: a master node may be a single point of failure.
 - To be **simple and lightweight**: tons of frames are sent, there must be no overhead.

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Channel Partitioning Protocols: TDMA

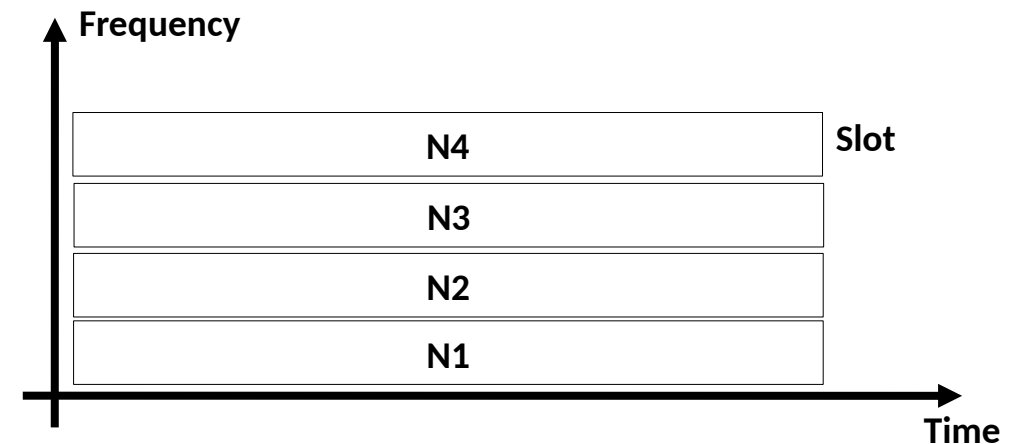
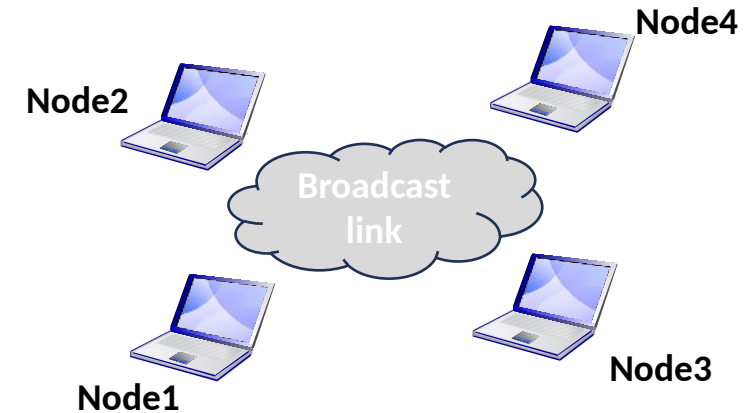
- **Time division multiple access (TDMA)**: let's consider a channel with N nodes having transmission rate of R bps, **TDMA divides time into time frames (steps) and further divides each time frame into N time slots.**
- Each **time slot is assigned to one** of the N nodes. Whenever a node has a packet to send, it waits for the assigned time slot.
- Typically, **slot sizes are chosen so that a whole packet can be transmitted** during a slot time.



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Channel Partitioning Protocols: FDMA

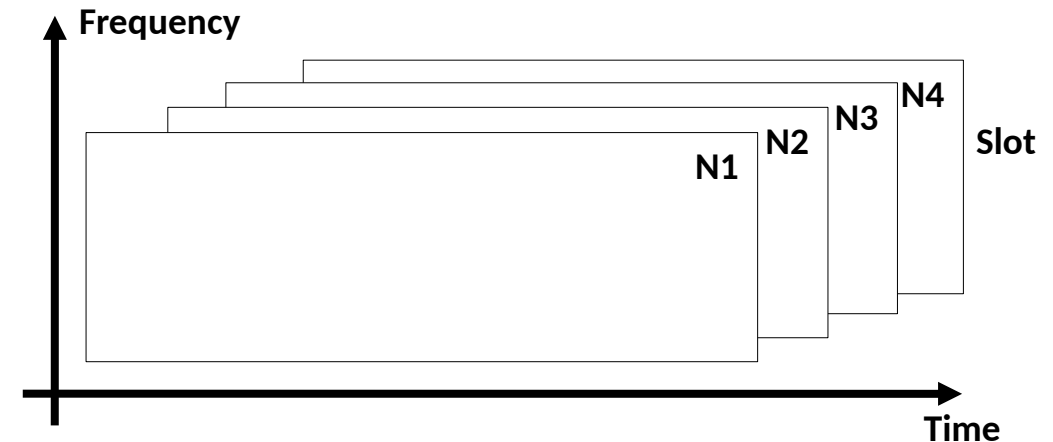
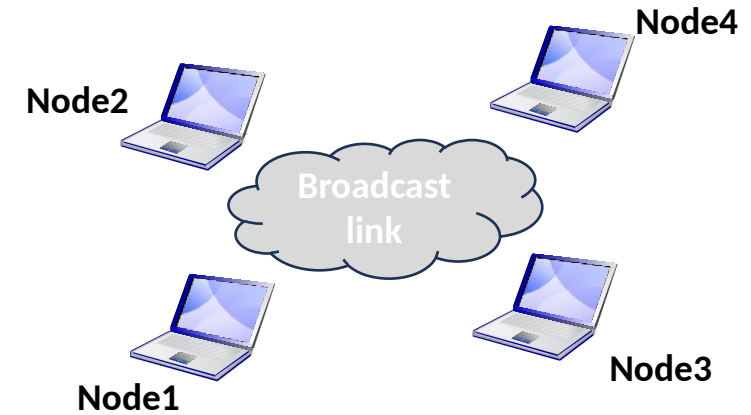
- **Frequency division multiple-access (FDMA)**: divides the R bps channel into different frequencies (each with a bandwidth of R/N) and assigns each frequency to one of the N nodes.
- FDMA and TDMA share pros and cons:
 - They **avoids collisions and divide the bandwidth fairly** among the N nodes.
 - A **node is limited to a bandwidth of R/N** , even when it is the only node with packets to send.



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Channel Partitioning Protocols: CDMA

- **Code division multiple access (CDMA)**: assigns a different code to each node which is used to encode/decode the data bits it sends.
- If the codes are chosen carefully, **CDMA networks allows different nodes to transmit simultaneously** without causing interference.
- CDMA works **mainly on wireless channels**; it has been used in military systems for some time (due to its anti-jamming properties) and **also in cellular telephony** (e.g., 3G).
 - Complex to implement.
 - It does not scale well with the number of nodes.



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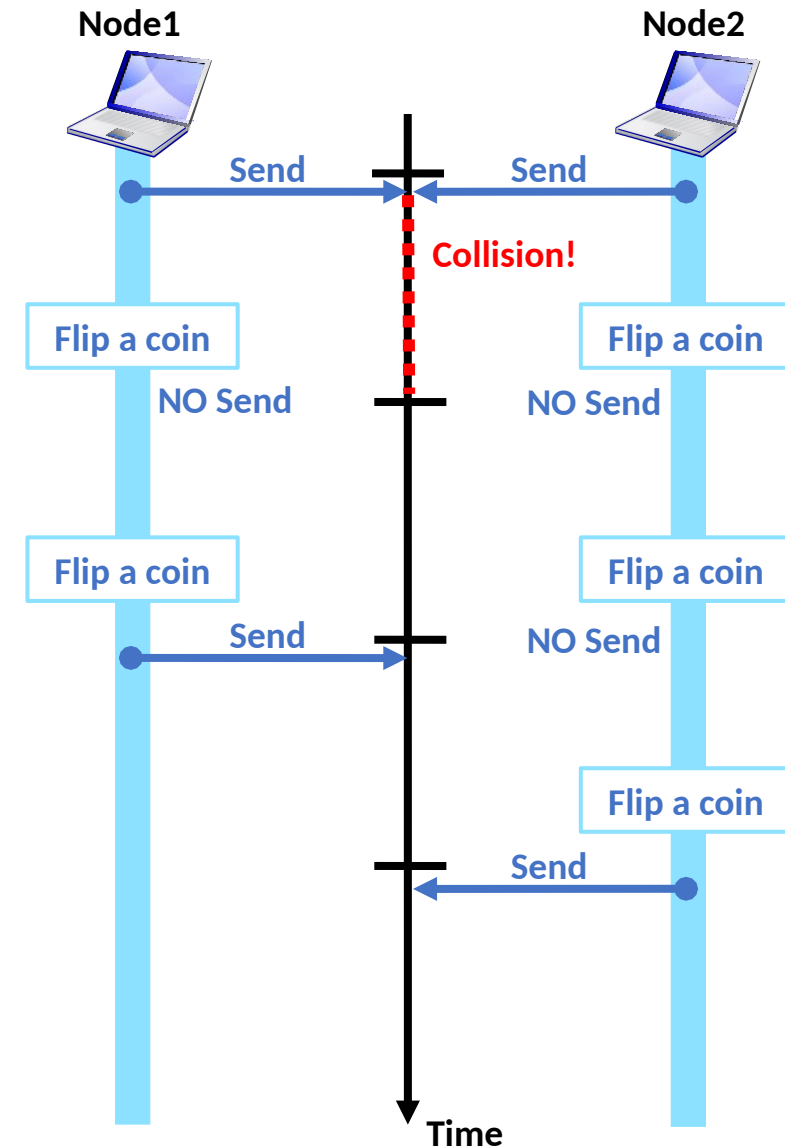
Random Access Protocols

- In random access protocols, a **transmitting node always transmits at the full rate** of the channel (at R bps).
- If a **collision occurs** (i.e., at least 2 nodes are transmitting) all transmitting **nodes waits a random delay** before attempting again the retransmission.
- Since this **selection is performed independently**, it is possible for the 2 nodes to **choose a delay which is different enough** to allow one of the two contenders to sneak the message in.
- If, otherwise, a **similar delay is chosen**, a new collision occurs, and **the process is iterated**.

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Random Access Protocols: Slotted ALOHA

- The **slotted ALOHA protocol** works as follows:
 - The **time is divided into slots** (as in TDMA) where each slot is large enough to contain one frame.
 - The nodes are synchronized, so **each node transmit frames only at the beginnings of a slots**.
 - **If no collision** is detected within the slot, the **communication continues**.
 - Otherwise, **if a collision is detected** within a slot, **each colliding node have a probability $p \in [0,1]$ of resending this frame** in each one of the following slots, until the frame is successfully sent.
- Features:
 - **If there is only one node**, it will use the **full rate R** of the channel (no collisions).
 - Is **decentralized** as nodes are totally independent from the others besides the synchronization.
 - Is **simple to implement** and to execute (random selection is quite fast).
 - Having **multiple consecutive collisions** is unlikely.



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Random Access Protocols: CSMA

- One **weakness of ALOHA** is that we may start transmitting (producing a collision) **even if the channel is already busy**. We understand it just through the effect of collisions.
- A solution is **to monitor the channel** and to attempt transmission only if the channel is idle.
- **Carrier sense multiple access (CSMA)** and **CSMA with collision detection (CSMA/CD)** protocols are based on 2 principles:
 - **Carrier sensing**: nodes **listen to the channel before transmitting**. If a frame from another node is currently being transmitted, it waits until no transmission is detected.
 - **Collision detection**: nodes **listens to the channel while it is transmitting**. If a collision is detected, it stops transmitting and waits a random amount of time before restarting.
- If all nodes perform carrier sensing, **why do collisions occur in the first place?**